



Series 54/74

DM7131/DM8131, DM7136/DM8136

DM7131/DM8131, DM7136/DM8136 6-bit unified bus comparator general description

The DM7131/DM8131, DM7136/DM8136 compare two binary words of two-to-six-bits in length and indicates matching bit-for-bit of the two words. Inputs for one word are 54/74 series-compatible TTL inputs, whereas those of the second word are high impedance receivers driven by a terminated data bus. These bus inputs include 1V typical hysteresis which provides 1.8V noise immunity. The DM7131/DM8131 has active pull up output and goes to the low state upon comparison. The DM7136/DM8136 has open-collector output which goes to high state upon comparison and is expandable to n bits by collector-ORing. Both devices have an output latch which is strobe controlled.

The transfer of information to the output occurs when the STROBE input goes from a logic "1"

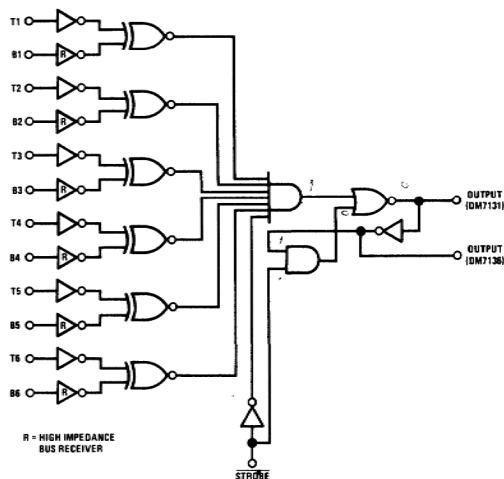
to logic "0" state. Inputs may be changed while the STROBE is at the logic "1" level without affecting the state of output. These devices are useful as address comparators in computer systems utilizing unified data bus organization.

features

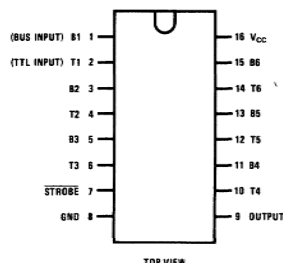
- Low bus input current 15 μ A typ
- High bus input noise immunity 1.8V typ
- High fan out
- Input clamping diodes
- Output compatible with TTL circuits
- Output latch provision

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logic and connection diagrams



Dual-In-Line and Flat Package



absolute maximum ratings

operating conditions

			MIN	TYP	MAX	UNITS
Supply Voltage	7V	Supply Voltage				
TTL and Strobe Input Voltage	5.5V	DM7131,DM7136	4.5	5	5.5	V
Fan Out	10	DM8131,DM8136	4.75	5	5.25	V
Storage Temperature Range	-65°C to +150°C	Temperature (T _A)				
		DM7131,DM7136	-55	+25	+125	°C
		DM8131,DM8136	0	+25	+70	°C

electrical characteristics (Operating Temperature Range — Unless Otherwise Specified)

PARAMETER	CONDITIONS	MIN	TYP	MAX	UNITS
Input "1" Voltage (Except Bus Inputs)	V _{CC} = Min	2.0			V
Input "0" Voltage (Except Bus Inputs)	V _{CC} = Min			0.8	V
Bus Input "1" Threshold Voltage	DM7131,DM7136	1.65	2.25	2.65	V
	DM8131,DM8136	1.80	2.25	2.50	V
Bus Input "0" Threshold Voltage	DM7131,DM7136	0.97	1.3	1.63	V
	DM8131,DM8136	1.05	1.3	1.55	V
Input Clamp Diode Voltage	V _{CC} = Max, I _{IN} = -12 mA T _A = +25°C			-1.5	V
TTL Input "1" Current	V _{CC} = Max, V _{IN} = 2.4V V _{IN} = 5.5V			40 1	μA mA
	V _{CC} = Max, V _{IN} = 0.4V			-1.6	mA
TTL Input "0" Current	V _{CC} = Max, V _{IN} = 2.4V V _{IN} = 5.5V			80 2	μA mA
	V _{CC} = Max, V _{IN} = 0.4V			-2.4	mA
Strobe Input "1" Current	V _{CC} = Max, V _{IN} = 4V V _{CC} = 0V, V _{IN} = 4V		15 1	50 50	μA mA
	V _{CC} = Max, V _{IN} = 0.4V			-2.4	mA
Maximum Bus Input Current	V _{CC} = Max, V _{IN} = 4V V _{CC} = 0V, V _{IN} = 4V		15 1	50 50	μA mA
Logic "0" Output Voltage	V _{CC} = Min, I _{OUT} = 16 mA			0.4	V
Logic "1" Output Voltage	V _{CC} = Min, I _{OUT} = -400 μA	2.4			V
Logic "1" Output Current	V _{CC} = Max, V _{OUT} = 5V			250	μA
	V _{CC} = Max, V _{OUT} = 0V	-20 -18		-55 -55	mA mA
Output Short Circuit Current	V _{CC} = Max		50	74	mA
Power Supply Current	V _{CC} = Max		50	74	mA
Propagation Delays					
TTL Input to Output, t _{pd1}	V _{CC} = 5V, T _A = 25°C		20		ns
TTL Input to Output, t _{pd0}	V _{CC} = 5V, T _A = 25°C		20		ns
Bus Input to Output, t _{pd1}	V _{CC} = 5V, T _A = 25°C		30		ns
Bus Input to Output, t _{pd0}	V _{CC} = 5V, T _A = 25°C		30		ns
Strobe Input to Output, t _{pd1}	V _{CC} = 5V, T _A = 25°C		20		ns
Strobe Input to Output, t _{pd0}	V _{CC} = 5V, T _A = 25°C		20		ns